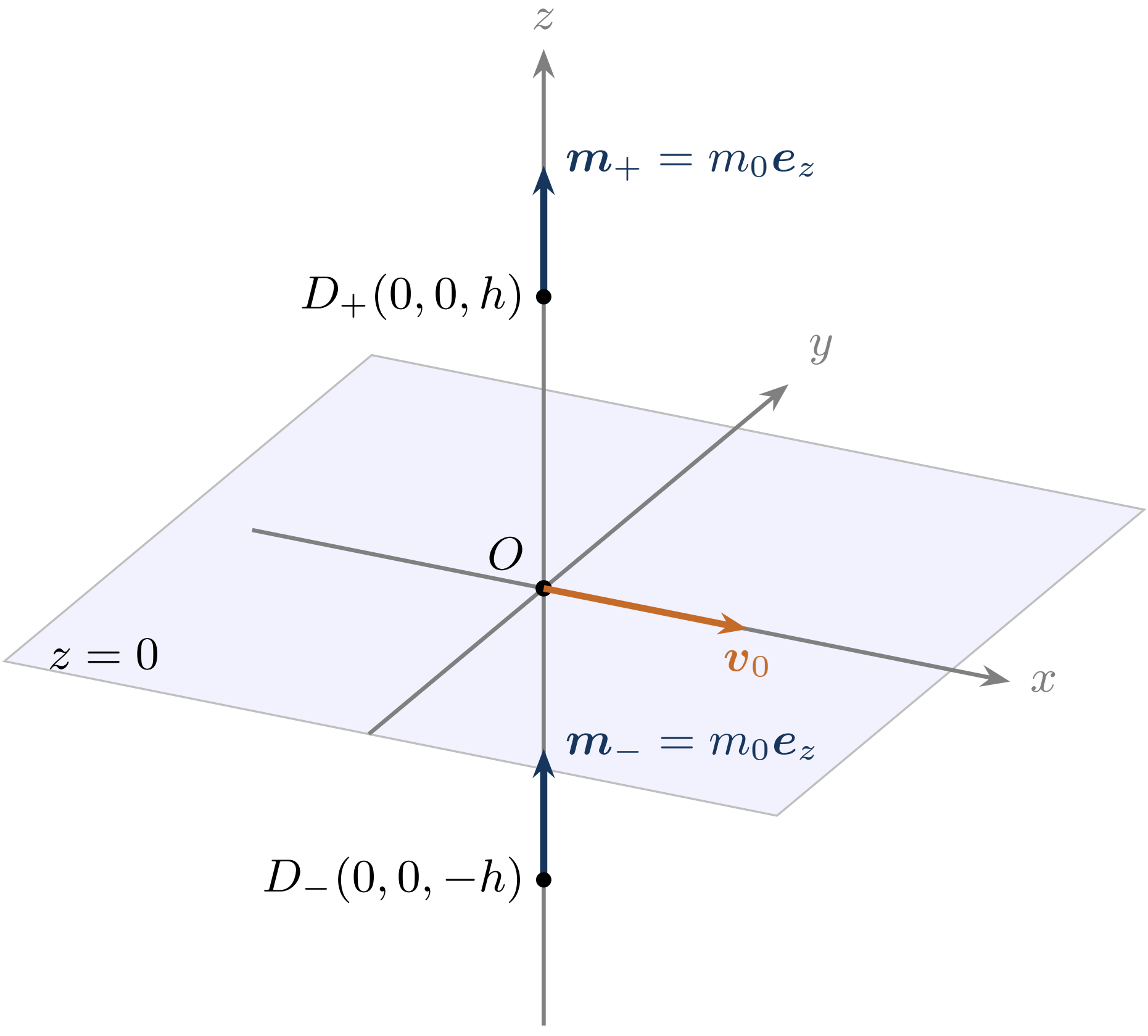
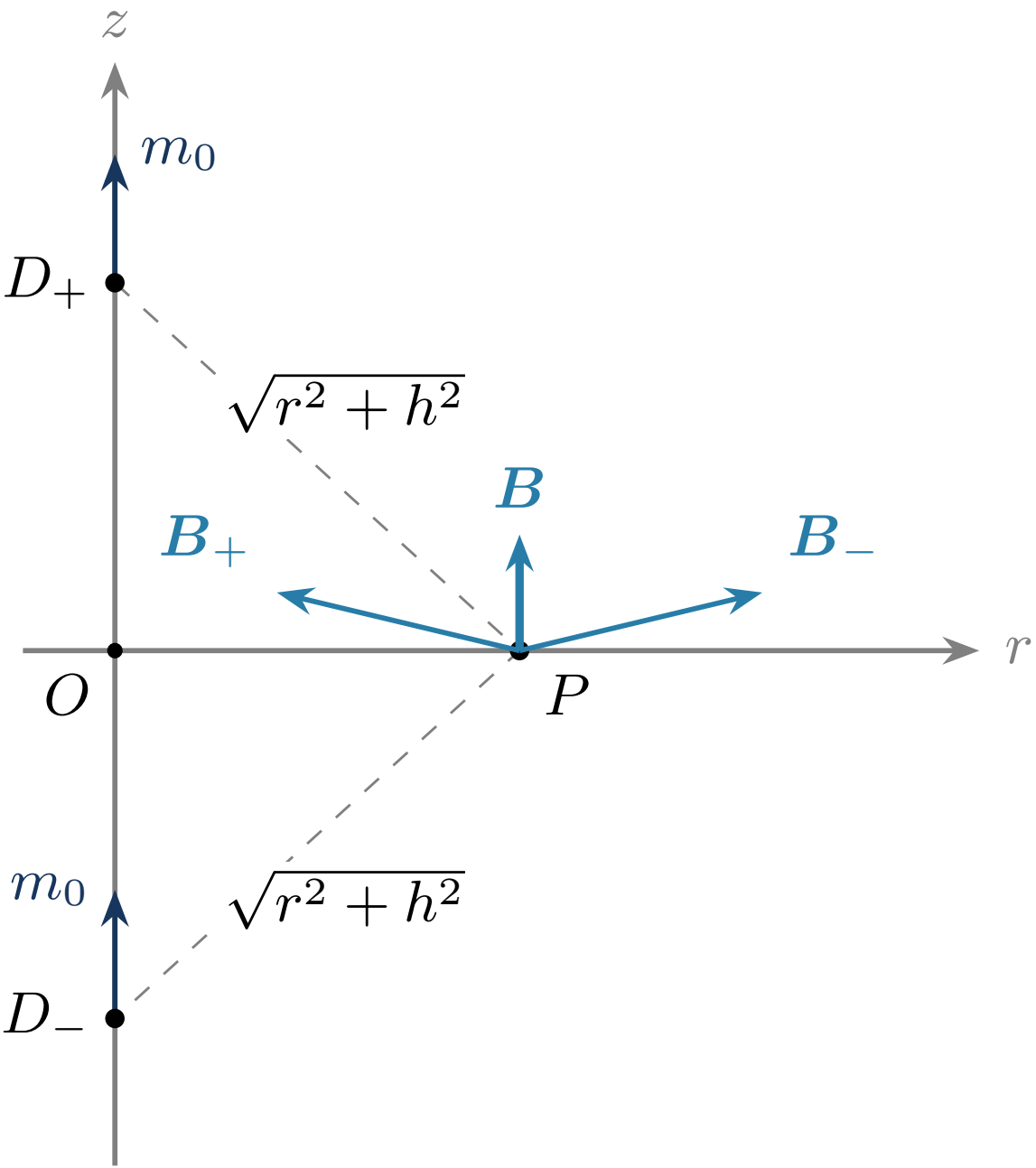

z
$\boldsymbol{m}_+ = m_0 \boldsymbol{e}_z$
$D_+(0,0,h)$
y
O
$z = 0$
$\boldsymbol{v}_0$
x
$\boldsymbol{m}_- = m_0 \boldsymbol{e}_z$
$D_-(0,0,-h)$



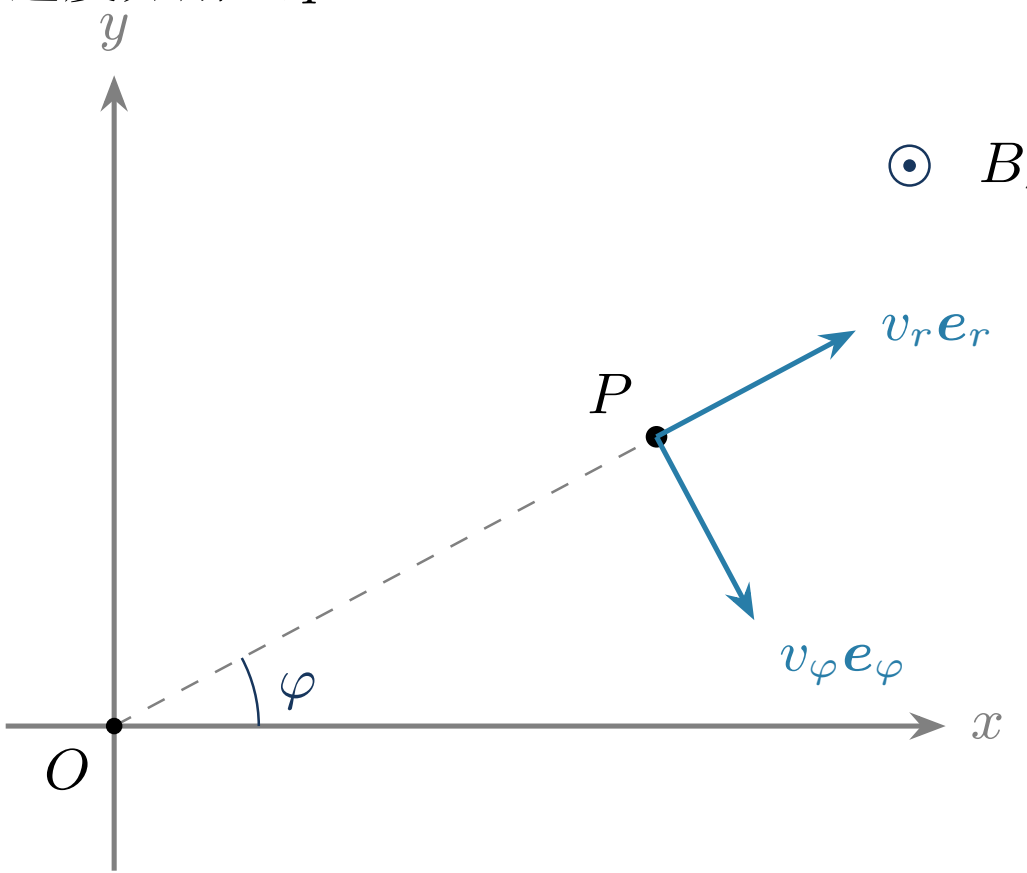
包含 z 轴与 P 的截面

两横向分量相消

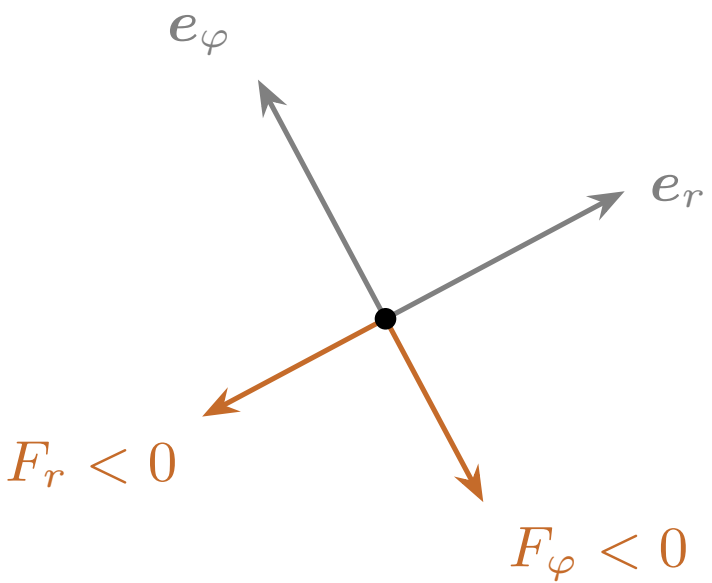
两轴向分量相加

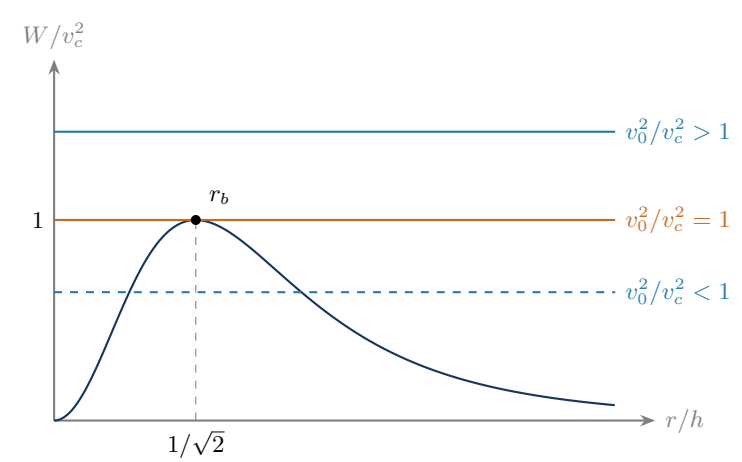
箭头示 $r < \sqrt{2}h$ 区域

速度分解（$q>0$、$v_r>0$、$B_z>0$）



同一位置的磁力分解





$\dot{r}^2 = v_0^2 - W(r)$

从原点出发需越过峰顶